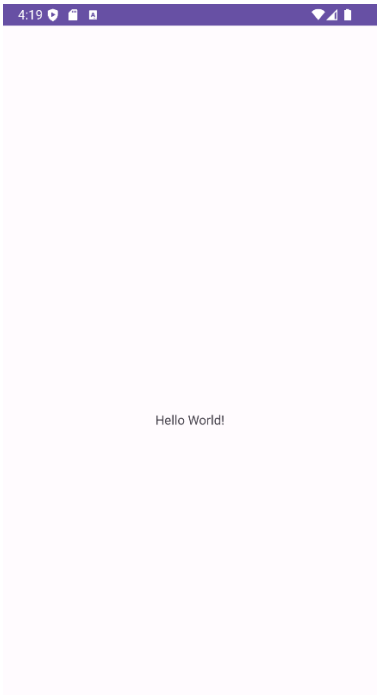
PROBLEM-SHEET-1

1. **create application that display "Hello World" in the middle of the screen in white color with black background**

activity_hello_world.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:background="#000000">

   <TextView
      android:id="@+id/helloText"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello World"
      android:textSize="28sp"
      android:textColor="#FFFFFF"
      android:layout_centerInParent="true"
      tools:ignore="HardcodedText" />

</RelativeLayout>
```

OUTPUT:



2. Create a login form with EditText fields for username and password. On button click, check the credentials and display a Toast message for a successful or failed login using setOnClickListener() ?

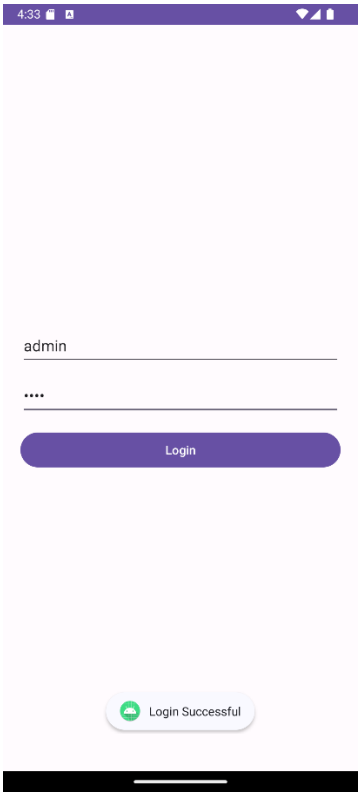
activity_login.html:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username"
        android:minHeight="48dp"
        android:textColorHint="#8D6E63"
        tools:ignore="HardcodedText,TextFields"
        android:autofillHints="" />

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        android:textColorHint="#8D6E63"
        tools:ignore="Autofill,HardcodedText" />

    <Button
        android:id="@+id/loginBtn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="15dp"
        tools:ignore="HardcodedText" />

</LinearLayout>
```

LoginActivity.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class LoginActivity extends AppCompatActivity {

    EditText username, password;
    Button loginBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        username = findViewById(R.id.username);
        password = findViewById(R.id.password);
        loginBtn = findViewById(R.id.loginBtn);

        loginBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                String user = username.getText().toString();
                String pass = password.getText().toString();

                if(user.equals("admin") && pass.equals("1234")) {
                    Toast.makeText(LoginActivity.this,
                            "Login Successful", Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(LoginActivity.this,
                            "Login Failed", Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

OUTPUT:



3. **Implement a button click listener that changes the button's background to black and the TextView's text color to white using setonclicklistener() only**

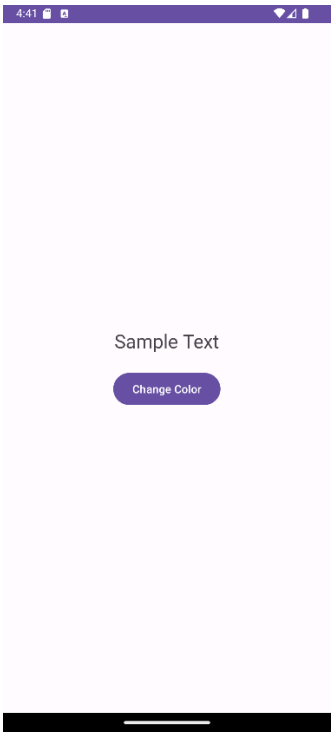
activity_question3.xml:

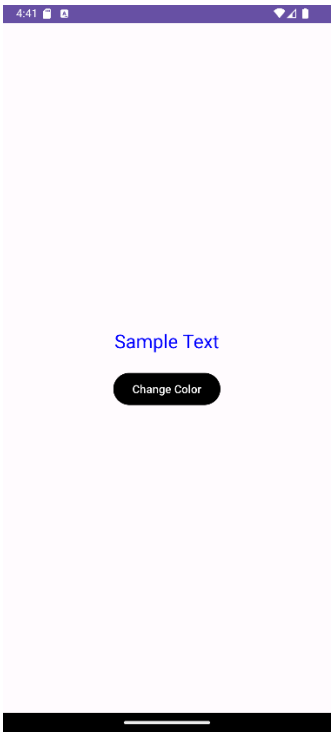
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Sample Text"
      android:textSize="24sp"
      android:layout_marginBottom="20dp"/>

   <Button
      android:id="@+id/changeBtn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Change Color"/> </LinearLayout>
```

activity_question3.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

public class activity_question3 extends AppCompatActivity {

    Button changeBtn;
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question3);

        changeBtn = findViewById(R.id.changeBtn);
        textView = findViewById(R.id.textView);

        changeBtn.setOnClickListener(v -> {
            changeBtn.setBackgroundColor(Color.BLACK);
            textView.setTextColor(Color.BLUE);
        });
    }
}
```

OUTPUT:





4. Build an app where users can pick multiple hobbies via CheckBoxes and display the selected hobbies in a Toast when a button is clicked, using setonclicklisterner() only.

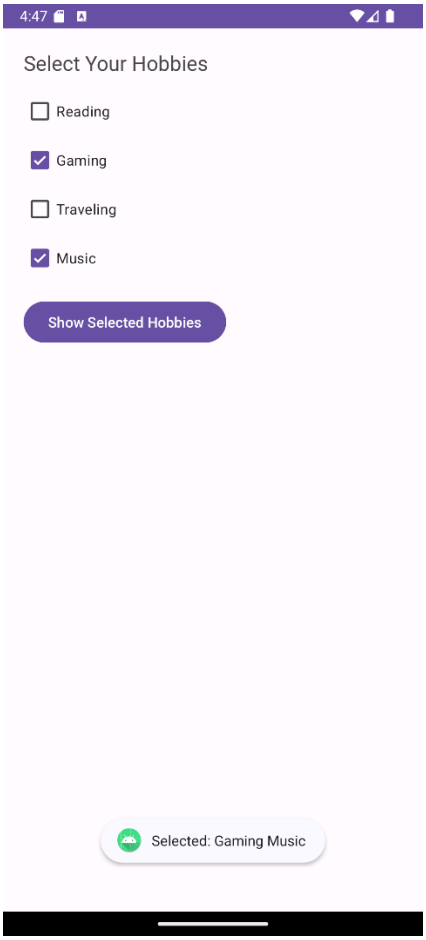
Activity_question4.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Select Your Hobbies"
      android:textSize="20sp"
      android:layout_marginBottom="10dp"/>

   <CheckBox
      android:id="@+id/cbReading"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Reading"/>

   <CheckBox
      android:id="@+id/cbGaming"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Gaming"/>

   <CheckBox
      android:id="@+id/cbTraveling"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Traveling"/>

   <CheckBox
      android:id="@+id/cbMusic"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Music"/>

   <Button
      android:id="@+id/showBtn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Show Selected Hobbies"
      android:layout_marginTop="15dp"/>

</LinearLayout>
```

activity_question4.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;

public class activity_question4 extends AppCompatActivity {

    CheckBox cbReading, cbGaming, cbTraveling, cbMusic;
    Button showBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question4);

        cbReading = findViewById(R.id.cbReading);
        cbGaming = findViewById(R.id.cbGaming);
        cbTraveling = findViewById(R.id.cbTraveling);
        cbMusic = findViewById(R.id.cbMusic);
        showBtn = findViewById(R.id.showBtn);

        showBtn.setOnClickListener(v -> {

            String hobbies = "";

            if(cbReading.isChecked())
                hobbies += "Reading ";

            if(cbGaming.isChecked())
                hobbies += "Gaming ";

            if(cbTraveling.isChecked())
                hobbies += "Traveling ";

            if(cbMusic.isChecked())
                hobbies += "Music ";

            Toast.makeText(activity_question4.this,
                    "Selected: " + hobbies,
                    Toast.LENGTH_SHORT).show();
        });
    }
}
```

OUTPUT:



5. Create an app with RadioButtons to choose a gender. Display selected value in a TextView.

Activity_question5.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Gender"
        android:textSize="20sp"
        android:layout_marginBottom="10dp"/>

    <RadioGroup
        android:id="@+id/radioGroupGender"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">
```

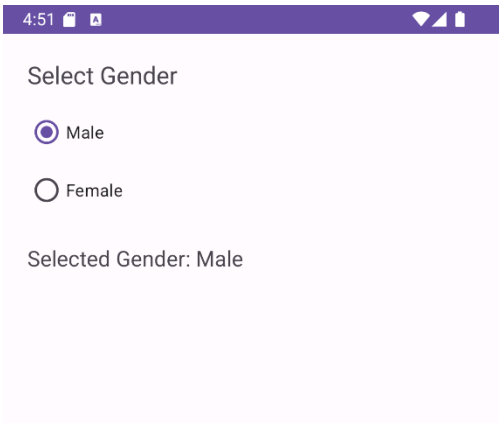
```
        <RadioButton
            android:id="@+id/radioMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male"/>

        <RadioButton
            android:id="@+id/radioFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female"/>

    </RadioGroup>

    <TextView
        android:id="@+id/resultText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Selected Gender: "
        android:textSize="18sp"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

activity_question5.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;

public class activity_question5 extends AppCompatActivity {

    RadioGroup radioGroupGender;
    TextView resultText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question5);

        radioGroupGender = findViewById(R.id.radioGroupGender);
        resultText = findViewById(R.id.resultText);

        radioGroupGender.setOnCheckedChangeListener((group, checkedId) -> {

            RadioButton selected = findViewById(checkedId);

            resultText.setText("Selected Gender: " + selected.getText());
```

```
        });
    }
}
```

OUTPUT:



6. **Create an app with a SeekBar and a TextView. Display the current value of the SeekBar in the TextView as the user slides it.**

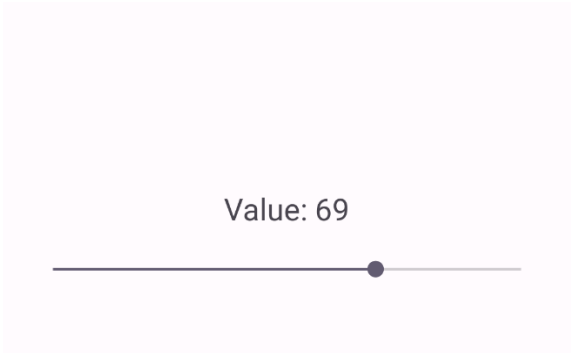
activity_question6.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/valueText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Value: 0"
        android:textSize="22sp"
        android:layout_marginBottom="20dp"/>

    <SeekBar
        android:id="@+id/seekBar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:max="100"/>

</LinearLayout>
```

activity_questions.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.SeekBar;
import android.widget.TextView;

public class activity_question6 extends AppCompatActivity {

    SeekBar seekBar;
    TextView valueText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question6);

        seekBar = findViewById(R.id.seekBar);
        valueText = findViewById(R.id.valueText);

        seekBar.setOnSeekBarChangeListener(new SeekBar.OnSeekBarChangeListener() {

            @Override
            public void onProgressChanged(SeekBar seekBar, int progress, boolean fromUser) {
                valueText.setText("Value: " + progress);
            }

            @Override
            public void onStartTrackingTouch(SeekBar seekBar) {}

            @Override
            public void onStopTrackingTouch(SeekBar seekBar) {}
        });
    }
}
```

OUTPUT:



7. Create an application to change screen color as per the user choice from a context menu

activity_question7.xml:

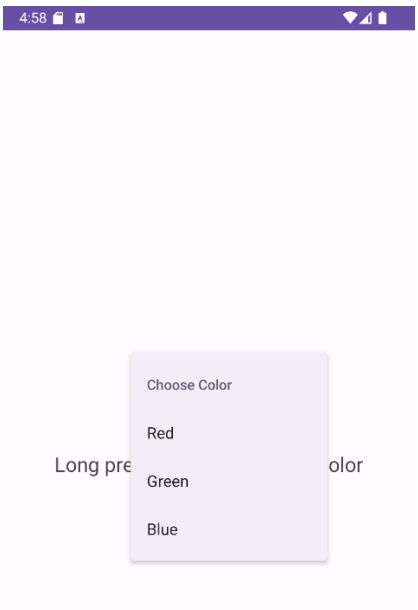
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/mainLayout"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical">

   <TextView
      android:id="@+id/textInfo"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Long press to change screen color"
      android:textSize="20sp"/>

</LinearLayout>
```

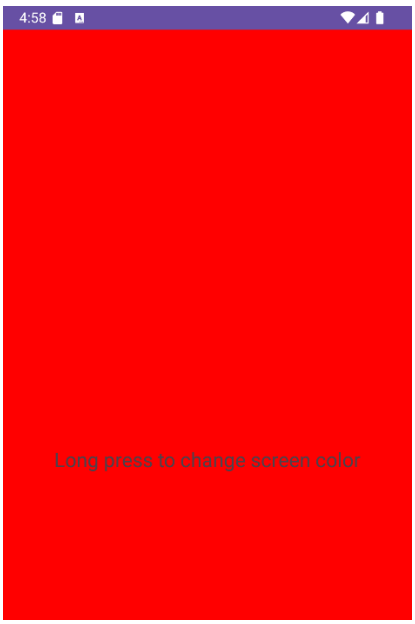
activity_question7.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
import android.os.Bundle;
import android.view.ContextMenu;
import android.view.MenuItem;
import android.view.View;
import android.widget.LinearLayout;

public class activity_question7 extends AppCompatActivity {

   LinearLayout mainLayout;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_question7);

      mainLayout = findViewById(R.id.mainLayout);

      registerForContextMenu(mainLayout);
   }

   @Override
   public void onCreateContextMenu(ContextMenu menu, View v,
                                   ContextMenu.ContextMenuInfo menuInfo) {
```

```
        super.onCreateContextMenu(menu, v, menuInfo);

        menu.setHeaderTitle("Choose Color");

        menu.add(0, 1, 0, "Red");
        menu.add(0, 2, 0, "Green");
        menu.add(0, 3, 0, "Blue");
    }

    @Override
    public boolean onContextItemSelected(MenuItem item) {

        switch (item.getItemId()) {

            case 1:
                mainLayout.setBackgroundColor(Color.RED);
                return true;

            case 2:
                mainLayout.setBackgroundColor(Color.GREEN);
                return true;

            case 3:
                mainLayout.setBackgroundColor(Color.BLUE);
                return true;

            default:
                return super.onContextItemSelected(item);
        }
    }
}
```

OUTPUT:





8. Develop an Android application using controls like Button, TextView, EditText for designinga calculator having basic functionality like Addition, Subtraction,Multiplication and Division.

activity_question8.xml:

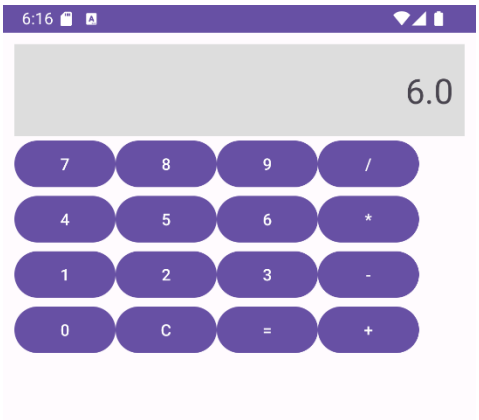
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="10dp">

   <TextView
      android:id="@+id/display"
      android:layout_width="match_parent"
      android:layout_height="80dp"
      android:text="0"
      android:textSize="30sp"
      android:gravity="right|center_vertical"
      android:background="#DDDDDD"
      android:padding="10dp"/>

   <GridLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:columnCount="4">

      <Button android:text="7" android:onClick="buttonClick"/>
      <Button android:text="8" android:onClick="buttonClick"/>
      <Button android:text="9" android:onClick="buttonClick"/>
      <Button android:text="/" android:onClick="buttonClick"/>

      <Button android:text="4" android:onClick="buttonClick"/>
      <Button android:text="5" android:onClick="buttonClick"/>
      <Button android:text="6" android:onClick="buttonClick"/>
      <Button android:text="*" android:onClick="buttonClick"/>

      <Button android:text="1" android:onClick="buttonClick"/>
      <Button android:text="2" android:onClick="buttonClick"/>
      <Button android:text="3" android:onClick="buttonClick"/>
      <Button android:text="-" android:onClick="buttonClick"/>

      <Button android:text="0" android:onClick="buttonClick"/>
      <Button android:text="C" android:onClick="buttonClick"/>
      <Button android:text="=" android:onClick="buttonClick"/>
      <Button android:text="+" android:onClick="buttonClick"/>

   </GridLayout>

</LinearLayout>
```

activity_question8.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class activity_question8 extends AppCompatActivity {

    TextView display;
    String current = "";
    double num1 = 0;
    String operator = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question8);

        display = findViewById(R.id.display);
    }

    public void buttonClick(View v) {

        Button btn = (Button) v;
        String text = btn.getText().toString();

        if(text.matches("[0-9]")) {
            current += text;
            display.setText(current);
        }

        else if(text.matches("[+\\-*/]")) {
            num1 = Double.parseDouble(current);
            operator = text;
            current = "";
        }

        else if(text.equals("=")) {

            double num2 = Double.parseDouble(current);
            double result = 0;

            switch(operator) {
                case "+": result = num1 + num2; break;
                case "-": result = num1 - num2; break;
                case "*": result = num1 * num2; break;
                case "/": result = num1 / num2; break;
```

```
            }

            display.setText(String.valueOf(result));
            current = String.valueOf(result);
        }

        else if(text.equals("C")) {
            current = "";
            display.setText("0");
        }
    }
}
```

OUTPUT:



9. **Create a custom view that draws a circle, a square, and a line using the Canvas and Paint classes, with all shapes outlined in red.**

activity_question9.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<com.example.androidproblemsheet.MyCustomView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

activity_question.java:

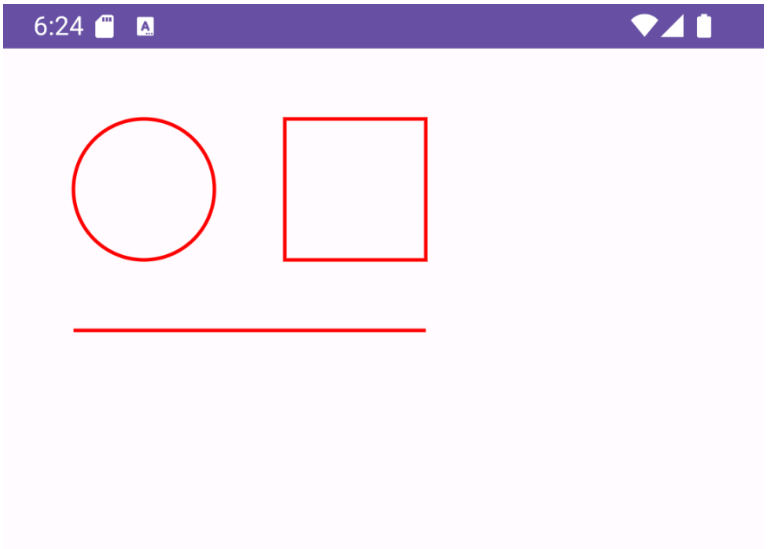
```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class activity_question9 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_question9);
    }
}
```

MyCustomView.java

```
package com.example.androidproblemsheet;

import android.content.Context;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.util.AttributeSet;
import android.view.View;

public class MyCustomView extends View {

    Paint paint;

    public MyCustomView(Context context) {
        super(context);
        init();
    }

    public MyCustomView(Context context, AttributeSet attrs) {
        super(context, attrs);
        init();
    }

    private void init() {
        paint = new Paint();
        paint.setColor(Color.RED);
        paint.setStyle(Paint.Style.STROKE);
        paint.setStrokeWidth(5);
    }

    @Override
    protected void onDraw(Canvas canvas) {
        super.onDraw(canvas);

        // Circle
        canvas.drawCircle(200, 200, 100, paint);

        // Square
        canvas.drawRect(400, 100, 600, 300, paint);

        // Line
        canvas.drawLine(100, 400, 600, 400, paint);
    }
}
```

OUTPUT:



10. Create an application to show Linear Gradient with Blue and White color inside a Circle

activity_question10.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<com.example.androidproblemsheet.GradientCircleView
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"/>
```

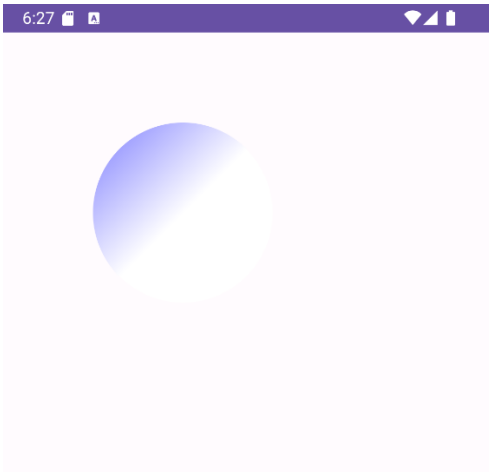
activity_question10.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class activity_question10 extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_question10);
   }
}
```

GradientCircleView.java

```
package com.example.androidproblemsheet;

import android.content.Context;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.LinearGradient;
import android.graphics.Paint;
import android.graphics.Shader;
import android.util.AttributeSet;
import android.view.View;

public class GradientCircleView extends View {

    Paint paint;

    public GradientCircleView(Context context) {
        super(context);
        init();
    }

    public GradientCircleView(Context context, AttributeSet attrs) {
        super(context, attrs);
        init();
    }

    private void init() {
        paint = new Paint();
    }

    @Override
    protected void onDraw(Canvas canvas) {
        super.onDraw(canvas);

        LinearGradient gradient = new LinearGradient(
                0, 0, 400, 400,
                Color.BLUE,
                Color.WHITE,
                Shader.TileMode.CLAMP
        );

        paint.setShader(gradient);

        canvas.drawCircle(400, 400, 200, paint);
    }
}
```

OUTPUT:



11. Set up two buttons—one to apply a fade-in animation to a text view on click and the other to apply a fade-out animation, handling both with setonclicklistener()

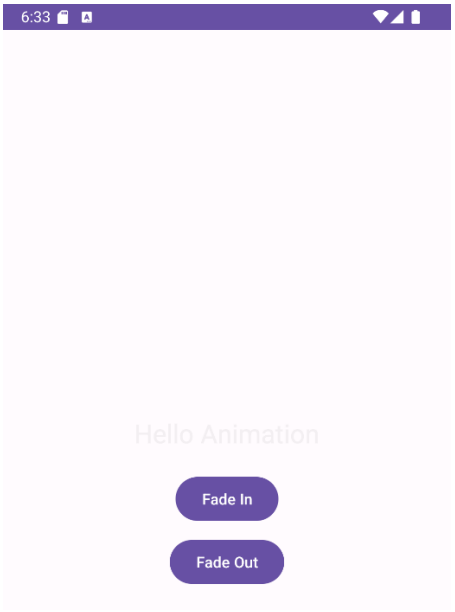
activity_question11.xml:

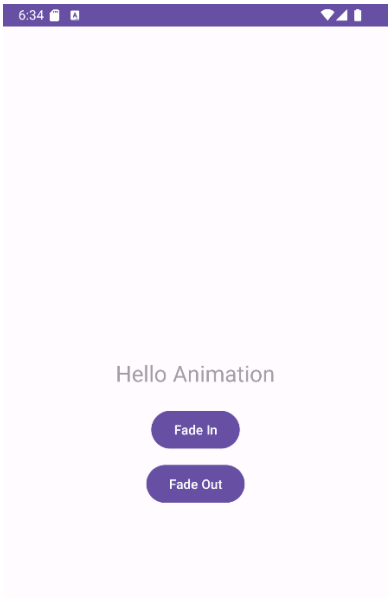
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Animation"
        android:textSize="24sp"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/fadeInBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fade In"/>

    <Button
        android:id="@+id/fadeOutBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fade Out"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

activity_question11.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.animation.AlphaAnimation;
import android.widget.Button;
import android.widget.TextView;

public class activity_question11 extends AppCompatActivity {

    TextView textView;
    Button fadeInBtn, fadeOutBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question11);

        textView = findViewById(R.id.textView);
        fadeInBtn = findViewById(R.id.fadeInBtn);
        fadeOutBtn = findViewById(R.id.fadeOutBtn);

        fadeInBtn.setOnClickListener(v -> {

            AlphaAnimation fadeIn = new AlphaAnimation(0.0f, 1.0f);
            fadeIn.setDuration(2000);
            textView.startAnimation(fadeIn);

        });

        fadeOutBtn.setOnClickListener(v -> {

            AlphaAnimation fadeOut = new AlphaAnimation(1.0f, 0.0f);
            fadeOut.setDuration(1000);
            textView.startAnimation(fadeOut);

        });
    }
}
```

OUTPUT:





12. Implement toggling between two activities using Intents

activity_question.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Activity 1"
      android:textSize="24sp"
      android:layout_marginBottom="20dp"/>

   <Button
      android:id="@+id/openSecondBtn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Go To Activity 2"/>

</LinearLayout>
```

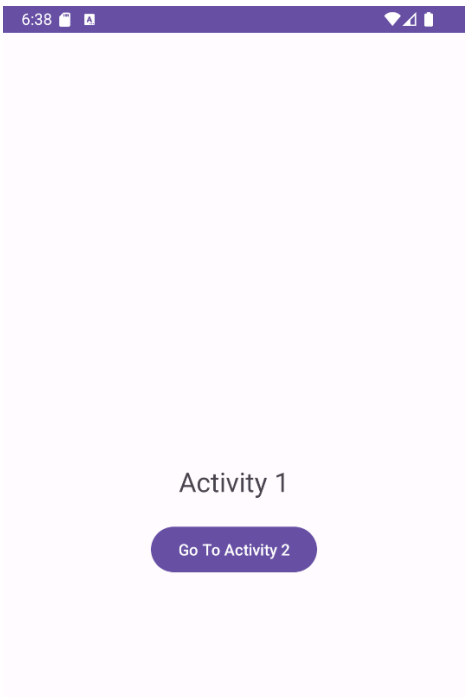
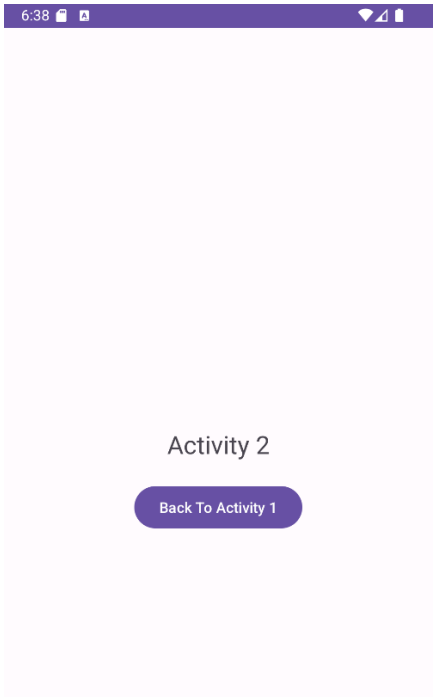
activity_question12.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

public class activity_question12 extends AppCompatActivity {

    Button openSecondBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question12);

        openSecondBtn = findViewById(R.id.openSecondBtn);

        openSecondBtn.setOnClickListener(v -> {
            Intent intent = new Intent(activity_question12.this, activity_question12_second.class);
            startActivity(intent);
        });
    }
}
```

activity_question12_second.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity 2"
        android:textSize="24sp"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/backBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Back To Activity 1"/>

</LinearLayout>
```

activity_question12_second.java:

```
package com.example.androidproblemsheet;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

public class activity_question12_second extends AppCompatActivity {

    Button backBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_question12_second);

        backBtn = findViewById(R.id.backBtn);

        backBtn.setOnClickListener(v -> {
            Intent intent = new Intent(activity_question12_second.this, activity_question12.class);
            startActivity(intent);
        });
    }
}
```

OUTPUT: